

Legacy Reuse Review - Betsybot-Software

Mapped against the DIY Robot Challenge SAD in `docs/diy-sad.html`.

What Can Be Reused Now

1. Motor / Drive Control

- Legacy drive package: `src/diy_motor_control_legacy`
- Source origin: Betsybot differential drive with Phoenix6 / TalonFX CAN control
- Useful for: baseline `/cmd_vel`-to-motor bridge, wheel odom, joint states
- Caveat: legacy differential-drive assumptions; should be wrapped by a safety gate and later replaced by STM32-centered control

2. Joystick Control

- Legacy launch pattern from `master_launch/launch/Joystick_drive.launch.py`
- Useful for: pre-competition manual driving and mapping
- Current bootstrap in this repo:
`src/challenge_bringup/launch/joystick_drive.launch.py`

3. Nav2 Bringup and Costmaps

- Legacy launch/config references from `betsybot/launch` and `betsybot/config/nav2_params.yaml`
- Useful for: Nav2 bringup structure, planner/controller wiring, RViz-friendly launch flow
- Current bootstrap in this repo:
`src/challenge_bringup/launch/challenge_master.launch.py` and `src/challenge_bringup/config/nav2_params.yaml`
- Updated for SAD: Hesai topic `/hesai/points`, Smac Hybrid planner, MPPI controller

4. Collision Monitor

- Legacy config: `betsybot/config/collision_monitor_params.yaml`
- Useful for: velocity safety layer around Nav2
- Current bootstrap in this repo:
`src/challenge_bringup/config/collision_monitor_params.yaml`

5. Sensors and Visualization

- Hesai driver source exists in the legacy workspace and was already buildable
- RealSense driver is standard `realsense2_ros` usage
- AprilTag detection stack is present for calibration/start-signal work

What Is Missing and Must Be Built

- E-stop controller and wireless fail-safe watchdog
- Command arbitration / mode mux for JOYSTICK, AUTONOMOUS, ESTOP_LOCK
- Mission state machine and obstacle behavior switcher
- PCL obstacle classifier and map differencing pipeline
- FAST-LIO2 runtime localization integration
- robot_localization EKF1/EKF2 and navsat_transform wiring
- Offline mapping pipeline using LIO-SAM

Step 1 Bootstrap Added Here

- Imported legacy drive package into `src/diy_motor_control_legacy`
- Created `challenge_bringup` package with SAD-aligned launch/config skeleton
- Added repo-local docs summary in `docs/reuse_plan_step1.md`

Immediate Next Development Tasks

1. Create `diy_cmd_vel_mux` to arbitrate joystick and autonomous control.
2. Add `diy_estop_controller` interface for hardware safety lockdown.
3. Wire in FAST-LIO2, EKF, and RTK localization next.